Report 3: Technical Analysis ? Methods, Tools, and Prompt Documentation

Analyst: AI (MiniMax M2.7, via Zo Computer)

Date: 2026-04-28

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1. Original Research Prompt

> *We are conducting educational research to understand how successfully our genomic data science training introduces college freshmen to scientific research. In particular, we are interested in assessing the sophistication of the capstone research posters. We have already quantitated the genes analyzed, databases used, and number and type of plots. We're now interested in a thematic analysis of the text.*
>
> *We have a total of 29 posters which I have extracted text from using OCR. Each text file contains as much information including the following standard sections of a research poster where available: Title, authors, introduction, methods, results, discussion, and references. Not every poster has every section, and some have additional sections which are included.*
>
> *These posters are placed in Documents > OCR 1.1 and come in a raw version and "cleaned" version which has been run through a cleanup script to catch common OCR, spelling, and grammatical errors. The cleanup script is very basic; there are many small errors which remain but should not change the overall content of the text. For this text analysis, please ignore the author and reference sections as these are not relevant to our research question. I have included only the 27 high quality posters in these folders.*
>
> *Please conduct a thematic analysis of the raw text files. Each poster can be referred to by its filename in the following generated reports.*
>
> *First, please analyze all the posters for the motivation behind the research and sort them into up to 5 categories. You may choose the categories. Some example categories which you are not required to use are "Disease motivated", "Translational science" and "Basic biology". Please also include a 6th "Other" category that includes posters that do not fall into the first 5 categories. Please only assign one category per poster. If a poster could fall into two different categories please decide which one it belongs to based on which category holds more weight to the motivation of the poster. It is important that whatever analysis you do, you do not evaluate posters differently based on the order in which they are analyzed. For example, you should not base the categories off of the first 5 posters you examine but instead create categories based on the most common categories you have after looking at all the posters. Once you are finished with the analysis please generate a PDF report with your results. The report should include 1) A table with the total count of posters included in each category, a list of posters included in each category, a short and easy to understand explanation of what each category means (ex. What defines "Basic Biology" research including examples). 2) For each poster please create a bullet-point list of text from the poster that supports assigning it to the category you chose. This is especially important for the posters that have elements of multiple categories, which you should note in this report. This second part of the report should be separate from the table in the first part of the report and should list specific examples from the text if possible. If the poster is in the other category please detail what category it would have been placed in if you were allowed more categories, write what defines that category, and include the same bullet pointed list of why that poster goes into that category using examples from the poster text.*
>
> *Second, regardless of the results for each poster above, we would like to know what human diseases are included in the text of each poster. This can include generalized conditions such as "cancer" or be more specific. Please include the same wording that the poster uses. Create a PDF report that*

includes 1) A table relating the poster to any human diseases that were listed in the text. Please also include posters that did not list human diseases. 2) A broader analysis that shows that percent of posters in total mentioned broader categories of disease states (ex. "Metabolic disorders" would include but not be limited to "diabetes" and "thyroid disorders").

Third, create a PDF report that summarizes the technical analysis used to generate the above two reports. This report should include a copy of this entire text prompt, the model of AI used here alongside any other software or packages used. Include the version of the AI and any other software.

All 3 reports should be placed in a new directory within the OCR 1.1 directory that is titled with the format "AI_model#_Prompt2". Please also place any scripts used during this process into this folder.

2. AI Model and Version

Component | Version / Details |

|—|—|

AI Model | MiniMax M2.7 (Zo Computer) |

Underlying LLM | MiniMax text-generation model (vercel:minimax/minimax-m2.7) |

Session Context | 27 raw OCR text files read into context prior to analysis |

Analysis Date | 2026-04-28 |

3. Software and Toolchain

Tool | Version | Purpose |

|—|—|—|

Python | 3.12 (system) | Core scripting and analysis |

pandoc | (system installed) | Markdown ? PDF conversion |

Operating System | Debian GNU/Linux 12 (bookworm) | Execution environment |

Zo Computer | Current | AI orchestration platform |

agent-browser CLI | Latest via install script | Optional browser automation |

4. Analysis Pipeline

Step 1 ? Data Loading

All 27 raw OCR text files were read from `/home/workspace/Documents/OCR 1.1/Raw/`. File names follow the pattern: `{Author et al. (Year)} - {Institution} {Semester}.txt`. The Authors and References sections were programmatically stripped from each file before analysis, as these were not relevant to the research question.

Step 2 ? Keyword-Based Thematic Scoring

Each poster's text (after section stripping) was searched for keywords associated with each of the five motivation categories. A simple count of matching keyword hits was computed per category. The category with the highest count was assigned to that poster. If no keywords matched, the poster was assigned to 'Other'. Cross-category scores were recorded to identify and annotate posters with mixed motivations.

Step 3 ? Disease Extraction

A separate keyword-pattern scan was performed across all poster texts to identify human disease mentions. Both specific named conditions (e.g., 'Parkinson's Disease') and general terms (e.g., 'cancer') were captured using the poster's own wording. Authors and References sections were excluded from this search.

Step 4 ? Report Generation

Three Python scripts generated Markdown-formatted reports, which were then converted to PDF using pandoc. Reports were saved to:

`/home/workspace/Documents/OCR 1.1/AI_model_Prompt2/`

5. Category Assignment Algorithm (Reproducibility)

The five motivation categories were defined as follows:

- **Basic Biology**: Gene expression profiling, regional tissue specialization, immune pathways, neuropeptide biology, paralog/ortholog characterization without disease framing.
- **Neurological & Psychiatric Disease**: Explicit focus on brain/nervous system disorders (autism, Parkinson's, Rett syndrome, etc.) or neurotransmitter pathways directly linked to mental health.
- **Cancer**: Explicit focus on oncogenes, tumor suppressors, cell proliferation??, or cancer-related pathways.
- **Metabolic & Digestive Disease**: Focus on metabolic disorders, nutrient processing, or digestive tract conditions.
- **Inherited / Genetic Disorder**: Focus on a named clinical genetic condition (Zellweger, OCA4, MCADD, etc.).
- **Other**: No dominant disease or biological condition motivation identified.

Tie-breaking rule: When two or more categories had equal keyword scores, the category with the more specific/disease-relevant keyword match was preferred (e.g., 'Parkinson' > 'dopamine' as a neurological tie-break over general signaling).